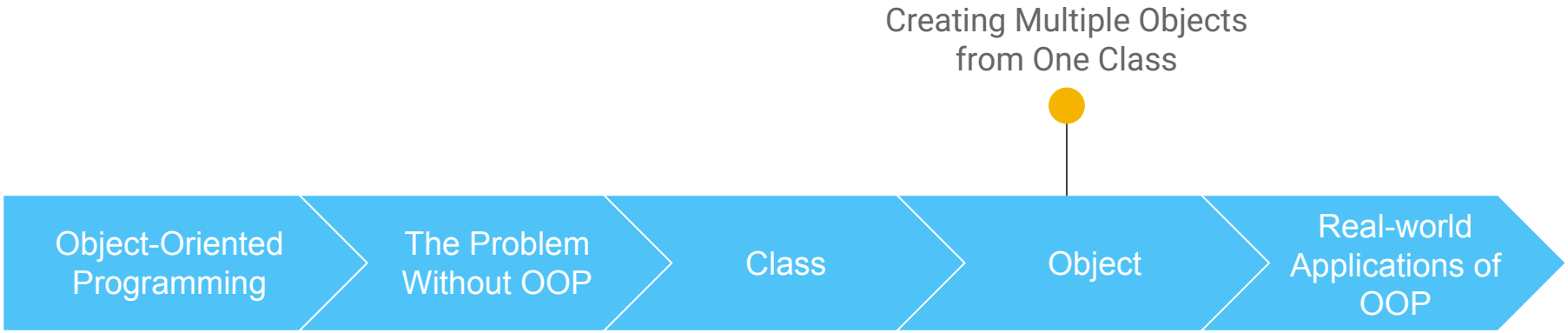
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WELCOME



Introduction to OOPS

Introduction to OOP



01

Object-Oriented Programming



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Object-Oriented Programming

Before OOP



- Let us think about a **Library**
- A library has many things **Books**, **Members**, **Shelves**, **Librarians etc..**

Object-Oriented Programming

Before OOP

Let us pick one thing:

A **Book**



Every book in the library has some **information**:

A Book

Title: Harry Potter

Author: J.K. Rowling

Available: Yes

Before OOP

And every book has some **actions** that can happen to it:

A Book

Information: Title, Author, Available

Actions: Borrow, Return, Display details



The **information** and the **actions**
both belong to the **same thing** (Book)

The Pattern

Phone

Information:

- Brand
- Model
- Battery %

Actions:

- Make a call
- Send message
- Take photo

Car

Information:

- Color
- Speed
- Fuel Level

Actions:

- Accelerate
- Brake
- Refuel

Student

Information:

- Name
- Roll Number
- Marks

Actions:

- Attend class
- Give exam
- View result

Object-Oriented Programming

The Pattern

Every real-world thing has:

Part	What it means	Examples
Information (Data)	What the thing **knows** about itself	Title, Color, Name, Battery %
Actions (Behaviour)	What the thing **can do**	Borrow, Accelerate, Send message



Both parts belong together

A car's speed belongs to that car and a phone's battery belongs to that phone

They are not floating around separately

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Object-Oriented Programming

OOP



OOP (Object-Oriented Programming) is a way of writing code that copies this real-world pattern

In OOP, we organise our code around **"things"** - just like the real world

Object-Oriented Programming

OOP



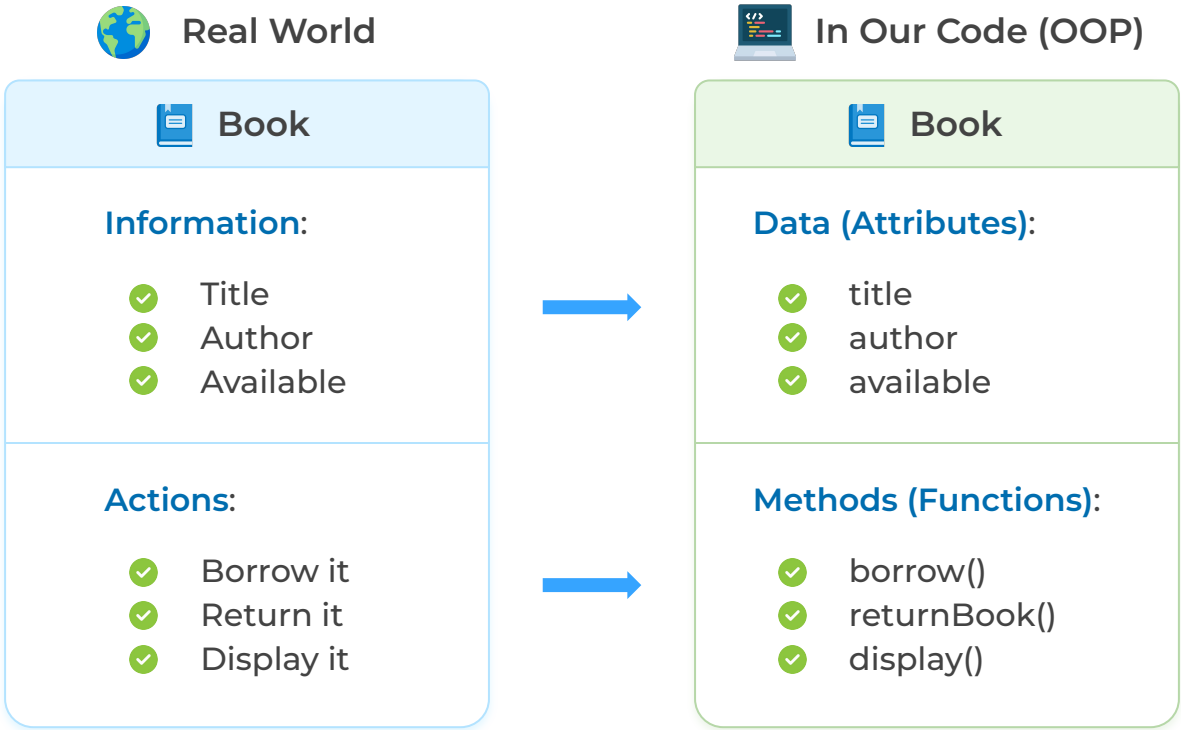
Each thing in our code keeps:

- Its own **information** (data)
- Its own **actions** (behaviour)

together in **one box**

Object-Oriented Programming

Real World → Code



Object-Oriented Programming

OOP



OOP is not JavaScript or Python or Java.

It is a **style of writing code**, a way of thinking

Object-Oriented Programming

OOP



Many languages support this style:



...and more

We will learn the **idea** in pseudocode first, then write it in JavaScript later

02

The Problem Without OOP



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The Problem Without OOP

What Happens Without OOP?

Now let us see what happens when we try to write our Library program **without** using this "**things in boxes**" approach

Pseudocode

```
bookTitle1 = "Harry Potter"
bookAuthor1 = "J.K. Rowling"
bookAvailable1 = true

bookTitle2 = "Atomic Habits"
bookAuthor2 = "James Clear"
bookAvailable2 = true
```



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The Problem Without OOP

What Happens Without OOP?

Now let us write a function to borrow a book:

Pseudocode

```
FUNCTION borrowBook(title):
  IF title == bookTitle1 AND bookAvailable1 == true:
      bookAvailable1 = false
      PRINT title + " borrowed successfully"
  ELSE IF title == bookTitle2 AND bookAvailable2 == true:
      bookAvailable2 = false
      PRINT title + " borrowed successfully"
  ELSE:
      PRINT "Book not available"
END FUNCTION
```



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The Problem Without OOP

What If We Add 100 Books?



bookTitle1, bookAuthor1, bookAvailable1

bookTitle2, bookAuthor2, bookAvailable2

bookTitle3, bookAuthor3, bookAvailable3 ...

bookTitle100, bookAuthor100, bookAvailable100

300 variables

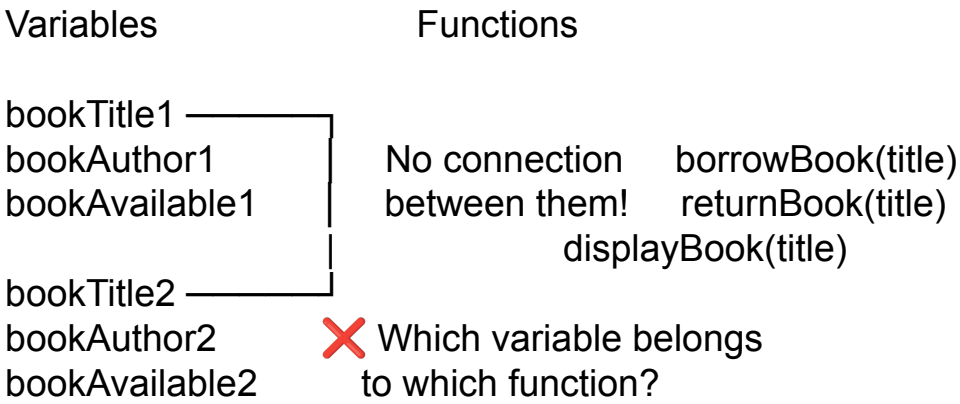
100 IF-ELSE conditions!

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The Problem Without OOP

What If We Add 100 Books?

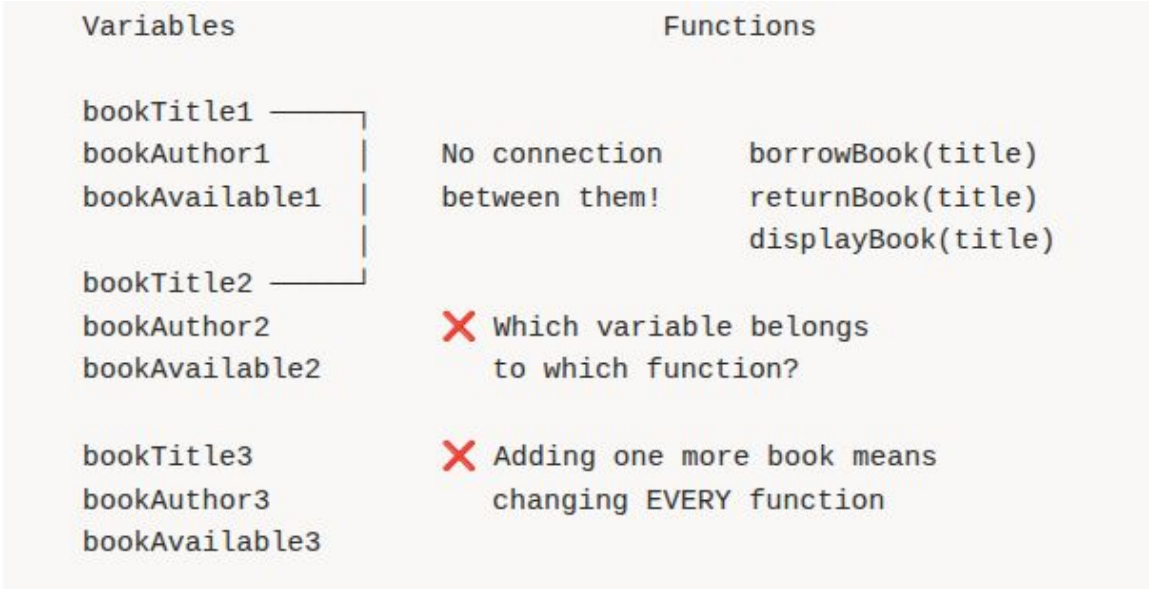
Visual: Why This Breaks



The Problem Without OOP

What If We Add 100 Books?

Visual: Why This Breaks



The Problem Without OOP

The Two Big Problems

Problem	What happens
Data is scattered	**bookTitle1** and **bookAuthor1** are separate variables, nothing tells us they belong to the same book
Functions are disconnected	**borrowBook()** does not know which group of variables is "one book", we manually connect them using IF-ELSE

- This way of writing step-by-step instructions with loose variables and functions is called **Procedural Programming**
- It works for small programs. It breaks when the program grows

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The Problem Without OOP

The Two Big Problems



PROCEDURAL (Loose pieces)

```
bookTitle1      = "Harry Potter"
bookAuthor1     = "Rowling"
bookAvailable1 = true
```

```
borrowBook(title)
returnBook(title)
displayBook(title)
```

- 100 books = 300 variables
- Every function checks every variable

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The Problem Without OOP

Solution



OOP (One box per thing)

Book

title = "Harry Potter"

author = "Rowling"

available = true

borrow()

returnBook()

display()

- One box = one book
- Data + actions live together
- 100 books = 100 boxes
- (no extra variables needed)

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Solution

Object Oriented Programming



But to create this box, we need two things:

- A **blueprint** that describes what every book will have
- A way to **create real books** from that blueprint

Solution

Object Oriented Programming

In OOP, we have specific names

Thing	OOP Name	Meaning
The blueprint	**Class**	Describes what data and actions a thing will have
The real thing	**Object**	A real instance created from the blueprint

03

Class
The Blueprint



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Class - The Blueprint

Class



A **Class** is a blueprint. It describes:

- What **data** (attributes) the thing will hold
- What **actions** (methods) the thing can do

A class does **not** hold actual values,
it only describes the **structure**

Class

Real-life Analogy: Architect's Floor Plan



FLOOR PLAN (Blueprint)

Rooms: 3 Bedrooms, 1 Kitchen, 2 Bathrooms

Features: Balcony, Parking

- ✗ We cannot live in a floor plan
- ✗ It is just a description on paper
- ✓ But we can **BUILD** many houses from it



Floor Plan = **Class** (the blueprint)

Real House = **Object** (the real thing built from it)

Class - The Blueprint

Writing Our First Class (Pseudocode)

Pseudocode

```
CLASS Book:
  ATTRIBUTES:
      title
      author
      available

  METHODS:
      borrow()
      returnBook()
      display()
END CLASS
```



Quiz Time!



04

Object
The Real Thing



Object - The Real Thing

Object



This is our blueprint, it says:

- Every Book will have a **title**, an **author**, and an **available** status
- Every Book can **borrow()**, **returnBook()**, and **display()**

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Object - The Real Thing

Object



But right now, **no actual book exists yet**.

We have only described the structure just like a floor plan on paper

Object - The Real Thing

Object



An Object is a real thing created from a class

When we create an object from a class, the object gets:

- Its **own copy** of the data (attributes)
- Access to the **actions** (methods) described in the class

Object - The Real Thing

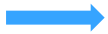
Object

Real-life Analogy (continued)

FLOOR PLAN (Class)

BLUEPRINT

- Rooms: 3
- Kitchen: 1
- Parking



REAL HOUSES (built from the plan - Object)

ALI'S HOUSE

- Rooms: 3
- Color: **Blue**
- Owner: Ali

ALI'S HOUSE

- Rooms: 3
- Color: **Red**
- Owner: Sam

- ➔ Cannot live here
- ➔ Just a description

- ➔ Ali's house
- ➔ A real house

- ➔ Sam's house
- ➔ A real house

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Object - The Real Thing

Object

Both houses were built from the **same plan**, but they are **independent**. Painting Ali's house blue does **not** make Sam's house blue

Object

Creating Objects

Pseudocode

```
CLASS Book:
  ATTRIBUTES:
      title
      author
      available
  METHODS:
      borrow()
      returnBook()
      display()
END CLASS
// Creating real books from the blueprint
book1 = NEW Book("Harry Potter", "J.K. Rowling", true)
book2 = NEW Book("Atomic Habits", "James Clear", true)
book3 = NEW Book("Clean Code", "Robert Martin", true)
```

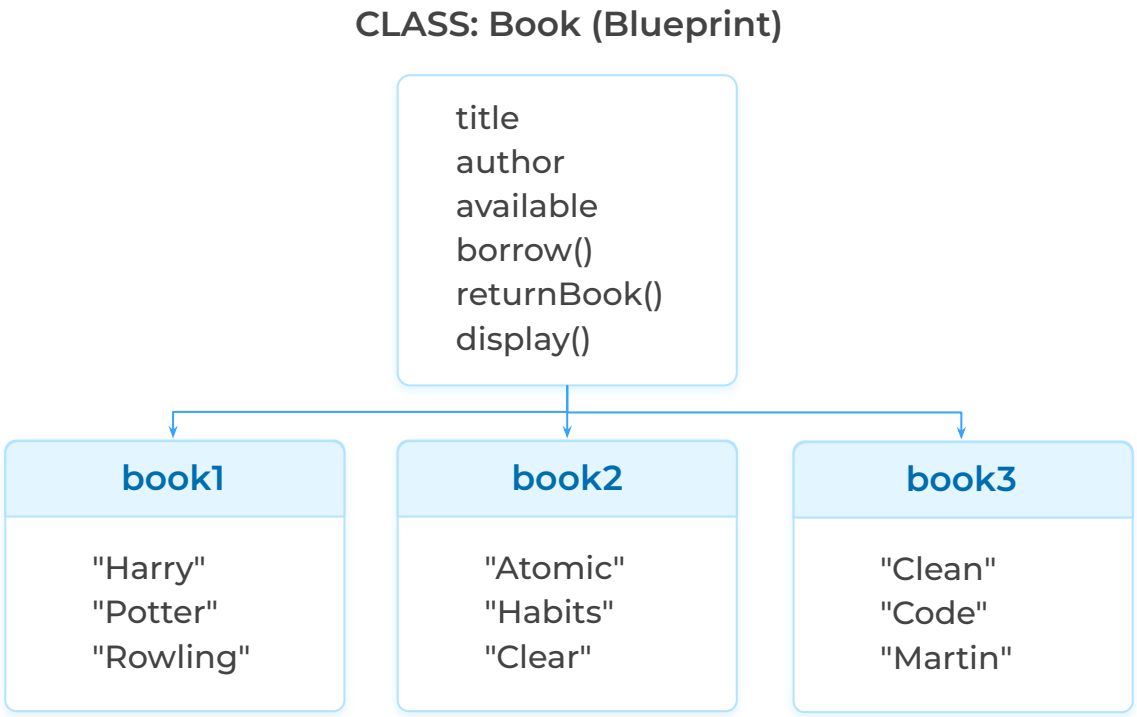
Now we have **three real books**, each with its own data



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Object

Creating Objects from-Book Class



Object

Class vs Object - The Key Difference

Class	Object
A blueprint / template	Describes what data and actions a thing will have
Describes the structure	Holds actual values
Created once	Can be created many times
Cannot be used directly	Can be used – we interact with objects
Example: The idea of a "Book"	Example: "Harry Potter" sitting on the shelf

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Object

Instantiation

Creating an object from a class is called **Instantiation**

The object is called an **Instance** of the class

```
book1 = NEW Book("Harry Potter", "J.K. Rowling", true)
```

This is INSTANTIATION book1 is an INSTANCE of Book

Class name (the blueprint)



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Object

Before OOP

Pseudocode

```
bookTitle1 = "Harry Potter"
bookAuthor1 = "J.K. Rowling"
bookAvailable1 = true


bookTitle2 = "Atomic Habits"
bookAuthor2 = "James Clear"
bookAvailable2 = true
```

Before OOP (Procedural)



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Object

Before OOP

Pseudocode

```
FUNCTION borrowBook(title):
  IF title == bookTitle1 AND bookAvailable1 == true:
      bookAvailable1 = false
  ELSE IF title == bookTitle2 AND bookAvailable2 == true:
      bookAvailable2 = false
  END IF
END FUNCTION


borrowBook("Harry Potter")
```

Before OOP (Procedural)



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Object

After OOP

Pseudocode

```
CLASS Book:
  ATTRIBUTES:
      title
      author
      available

  METHOD borrow():
      IF this.available == true:
          this.available = false
          PRINT this.title + " borrowed successfully"
      ELSE:
          PRINT this.title + " is not available"
      END IF
  END METHOD
END CLASS
```



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Object

After OOP

Pseudocode

```
book1 = NEW Book("Harry Potter", "J.K. Rowling", true)
book2 = NEW Book("Atomic Habits", "James Clear", true)

book1.borrow()
book2.borrow()
```



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05

Real-world Applications of OOP



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Real-world Applications of OOP

Where is OOP Used in Real Industry?

OOP is not just for learning. It is used everywhere in the software industry

Application	How OOP is Used
E-commerce (Amazon, Flipkart)	Describes what data and actions a thing will have
Social Media (Instagram, Twitter)	A real instance created from the blueprint
Banking Apps	Account, Transaction, Customer – objects protect sensitive data

Real-world Applications of OOP

Where is OOP Used in Real Industry?

Application	How OOP is Used
Games	Player, Enemy, Weapon, Level – game objects are reusable across levels
Our Library project	Book, Member, Library – we will build this across all 7 sessions

When we go for a placement interview, interviewers expect us to think in objects - not in loose variables and functions

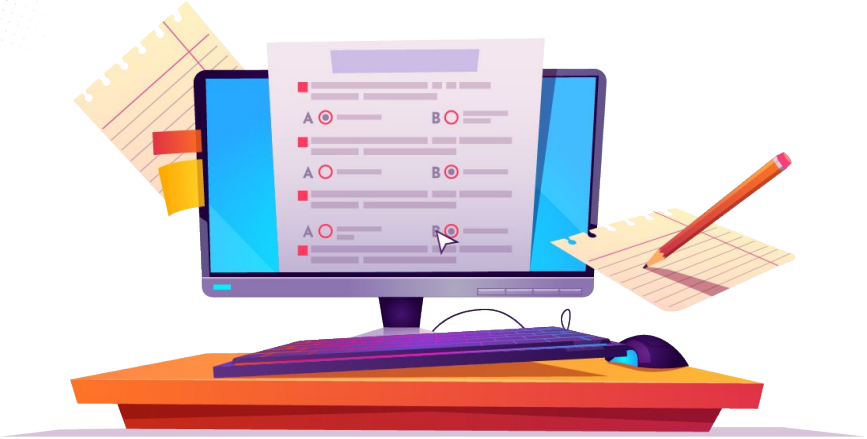
Quiz Time!



Key Takeaways

- Object Oriented Programming
- The Problem Without OOP
- Class
- Object
    - Creating Multiple Objects from One Class
- Real-world Applications of OOP

Practice



Feedback



Next Session

State, Behaviour & Constructor

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THANK YOU

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ALL THE BEST